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SHORT COMMUNICATION

EFFECT OF HERBAL DRUG SOMINA ON NEUROTRANSMITTER (DOPAMINE)

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ABSTRACT

The effects of Somina at the dose of 285-mg/kg were studied on the dopamine level in rat brain. Somina resulted in a significant increase in dopamine content. The increased concentrations of the neurotransmitters Dopamine in rat brain might be responsible to improve memory processes and reduces neuropsychiatric disorder.

INTRODUCTION

Stress and Anxiety can lead to mental and physical ill health (e.g. depression, nervous breakdown, and heart disease). Over 19 Million people in the United States (Borne, 2003), 20% women and 14% men in England (Mental Health Foundation, 2003) and 65% in Pakistan (Iqbal et al., 2002) experience anxiety or depression every day. During the last decade the use of herbal medicines have been tried to alleviate the symptoms of depressive disorders using animal research (Sharma, 1993). One of medicine is Somina prepared by Hamdard Laboratories (Waqf) Pakistan. It is composed of five medicinal plants: Sesamum indicum (14%), Prunus amygdalus (12%), Papaver somniferum (10%), Lactuca scariola (5%), and Lagenaria vulgaris (12%). It is claimed that Somina is a cephalic tonic having sedative, hypnotic, anxiolytic activities. It is prescribed in weakness of brain and memory. It is assumed that Somina might facilitate the neurotransmitters (such as Dopamine, Catecholamines etc.) presumed to play a vital role in the organization of brain functions. Therefore, the present work has been planned to study the effect of Somina used in the treatment of mental illness on neuro-transmitters (Dopamine).

MATERIALS AND METHODS

Sprague-Dawley rats of either sex weighing 200-225g were used in the present study. They were obtained from animal house of Dr. HMI IPHS, Hamdard University. Animals were housed in-groups of two (fourteen in each), and had free access to food and water adlibitum. Group I served Control and Group II as treated. Somina was obtained from Hamdard Laboratories (Waqf) Pakistan. The dose of Somina 285-mg/kg was prepared by dissolving its powder in distilled water and administered orally in rats for 29 consecutive days. While the control group received distilled water via the same route. At the end of 29th day all animals were sacrificed by decapitation that conforms the ethical standard and their brains were dissected out and stored at –70^{o}C till analysis. Concentrations of brain Dopamine and dihydroxyphenylacetic acid (DOPAC) were measured by using HPLC-EC as described earlier (Haider and Haleem, 2000). Statistical analysis was carried out by student's t-test. The criteria of significance was set at P<0.05.

RESULTS

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The results presented in Fig. 1 demonstrate that after Somina treatment brain Dopamine levels were found to increase significantly (P<0.05). However, DOPAC was

Fig. 1

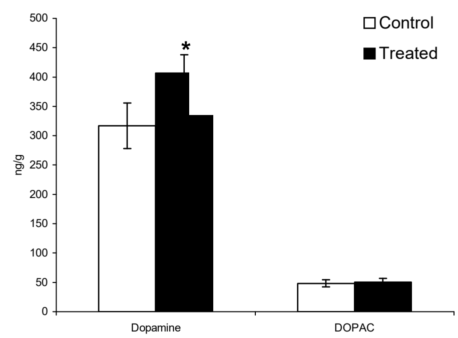

Fig.1: Effect of Somina on Brain Dopamine and DOPAC. Values are mean $\pm$ SE (n=14). Significant Difference by Students t-test. *P<0.05 from respective control.

also found to increase nonsignificantly (P>0.05) when compared to their respective controls (Fig. 1).

DISCUSSION

In the present study Somina has been tested for its effects on neurotransmitters. The results presented (Fig. 1) clearly shows that Somina drug increases the Dopamine levels in rats brain. Dopamine plays important roles in cognitive functions (Berman and Weinberger 1990; Luciana and Collins 1997). It is reported that decreased dopamine may lead to the development of catalepsy and depression in the animal (Marin, 1980; Fuenmayer and Vogt, 1979) and impaired working memory (Mizoguchi et al., 2000). Meschler et al (2000) suggested that Dopamine receptor agonist potentiate sedation. In the present study, it is therefore concluded that administration of Somina play a vital role in the organization of memory and decrease the chances of neuropsychiatric disorders and act as a sedative and anxiolytic by increasing levels of Dopamine in brain.

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